2018 JC2 H1 BIO P2 PRELIM EXAM ANSWER SCHEME

Section A

Answer **all** the questions in this section.

1 Fig. 1.1 shows part of a eukaryotic cell.

Fig. 1.1

(a) **(i)** Outline the role of organelle labelled **A**.

1. link rxn in matrix

pdc CO_2 & NADH where pyruvate undergoes oxidative decarboxylation;

2. Krebs cycle in matrix

pdc CO_2 & NADH where acetyl-coA undergoes oxidative decarboxylation;

3. oxidative phosphorylation on inner memb

pdc ATP via chemiosmosis; [3]

(ii) Identify **one** molecule with different mode of transport across the membrane of **A** and explain the mode of transport.

1. CO_2/ O_2;

tpted across memb via simple diffusion down conc. grad due to its small size & uncharged mol; OR

2. ATP/ ADP;

tpted across memb via facilitated diffusion as it is large in size & charged which is repelled by hydrophobic core of phospholipid bilayer; OR

3. pyruvate;

tpted across memb via facilitated diffusion as it is large in size & charged which is repelled by hydrophobic core of phospholipid bilayer; [2]

(b) **(i)** Explain the significance of glycolysis in aerobic respiration.

1. oxidation of glucose into pyruvate

via substrate level phosphorylation producing ATP;

2. NAD is reduced into NADH

NADH is supplied to inner memb of mito for oxidation phosphorylation;

3. pyruvate enters into matrix of mito undergoes oxidative decarboxylation

converted into acetyl coA for subsequent Krebs cycle to take place; [3]

An experiment was carried out to investigate the effect of temperature on respiration in isolated mitochondria extracted from a worm. Respiratory substrate was provided and oxygen consumption was monitored at 15°C, 25°C and 35°C.

Fig. 1.2 shows the temperature coefficients, Q_{10}, when temperature is increased from 15°C to 25°C and from 25°C to 35°C.

Q_{10} measures the ratio of the rate of respiration when the temperature increases by 10°C.

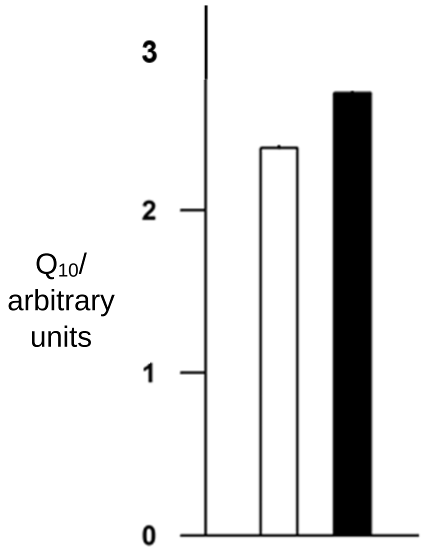

Fig. 1.2

(ii) Describe and explain the effect of temperature on the Q_{10} of mitochondria respiration.

1. increase in Q_{10} is higher for increase in temp from 25-35°C c.f. to 15-25°C

2.4 arbitrary units for 15-25°C compared to 2.7 arbitrary units for 25-35°C;

2. KE of both respiratory enz & substrate increases causing higher freq of effective collisions b/w them

more ES complexes formed, increasing rate of respiration; [2]

[Total: 10]

2 Fig. 2.1 is a photomicrograph of plant root cells near the growing top. Some of the cells are undergoing mitosis.

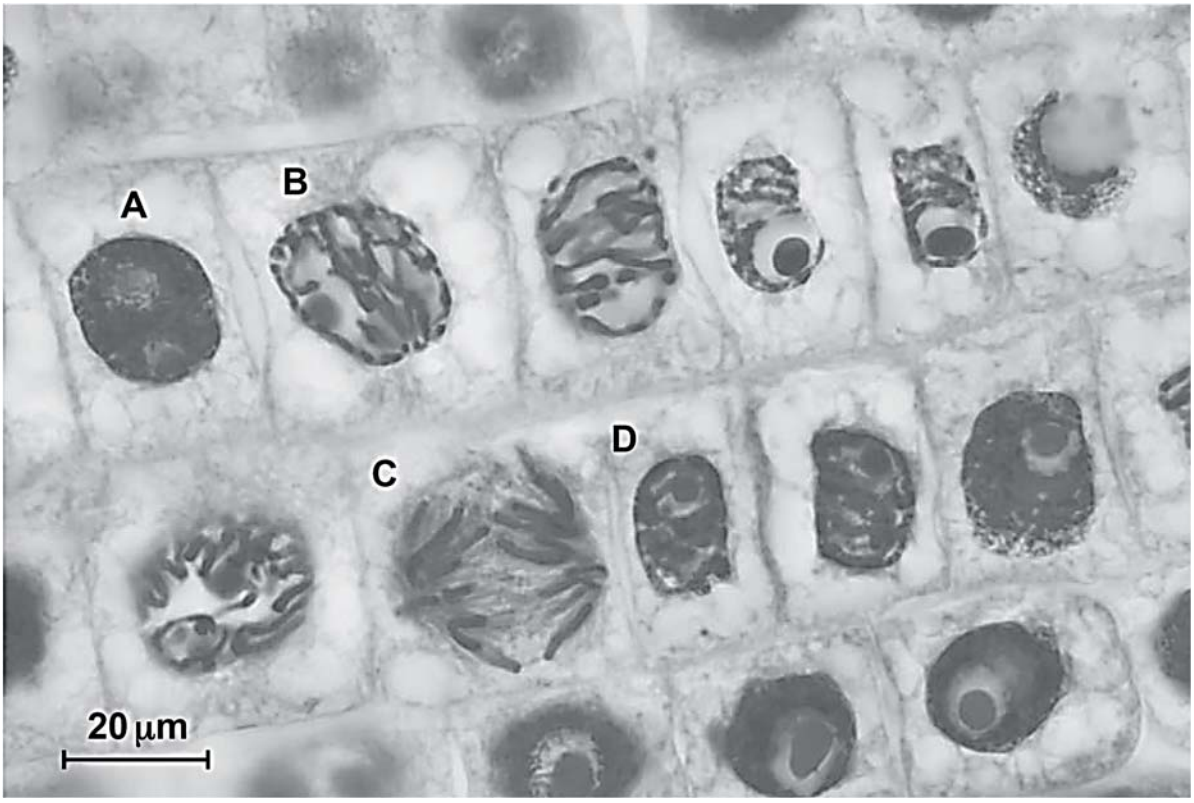

Fig. 2.1

(a) State **one** feature, visible in Fig. 2.1, which indicates that the section is taken from a plant tissue and not animal tissue.

presence of cell walls/ rectangular cells; [1]

(b) State the letter, **A** to **D**, of the cell in Fig. 2.1 which is in:

(i) prophase ***B;***

(ii) anaphase ***C;*** [2]

(c) Explain why the growth of roots involves mitosis and not meiosis.

1. mitosis maintain diploidy & pdc genetically identical daughter cells

results in genetic stability;

2. meiosis results in haploid cells & pdc genetically non-identical daughter cells

results in genetic variation;

3. growth of tissue requires similar cells to be grouped together

to perform same fn; [3]

(d) Fig. 2.2 represents one complete cell cycle for the plant root cells.

(i) Complete Fig. 2.2 by naming the stages represented by **J**, **K** and **L**.

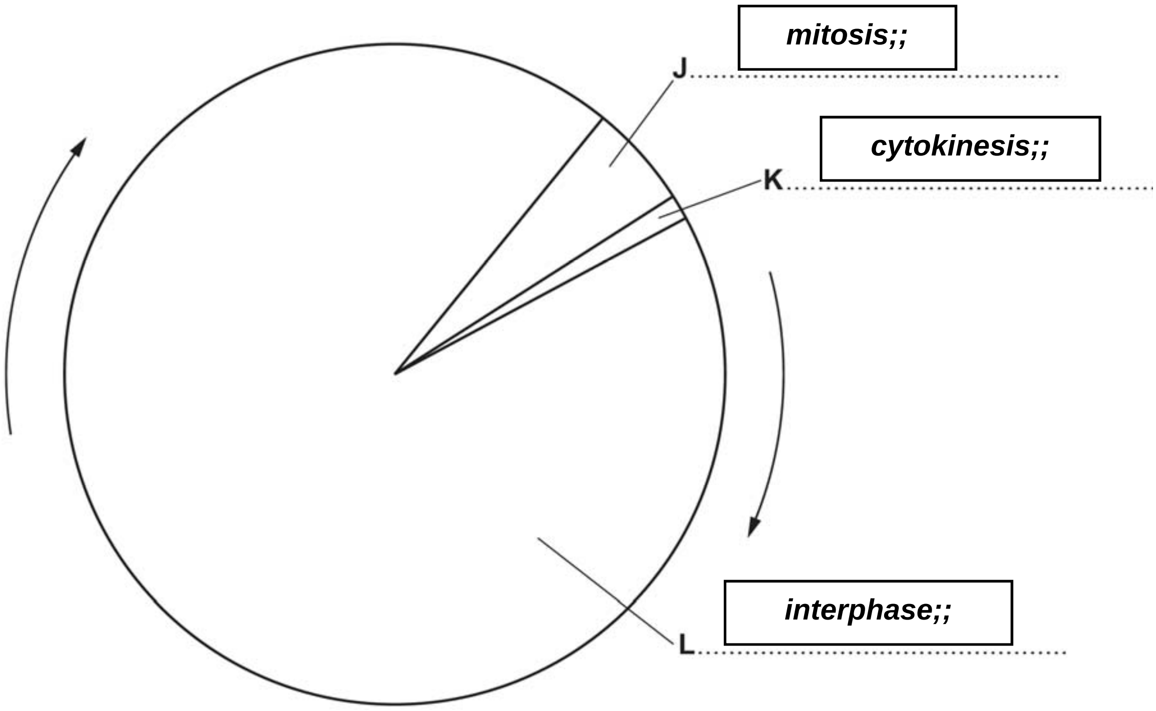

Fig. 2.2 [3]

(ii) Some bacteria infect the plant to result in tumour formation at the infected tissues as shown in Fig. 2.3.

Research on the tumour formation processes revealed that there was an unusual excessive production of auxin, an important plant growth hormone. Auxin has been found to regulate gene expression resulting in cell division, expansion, differentiation and specialisation.

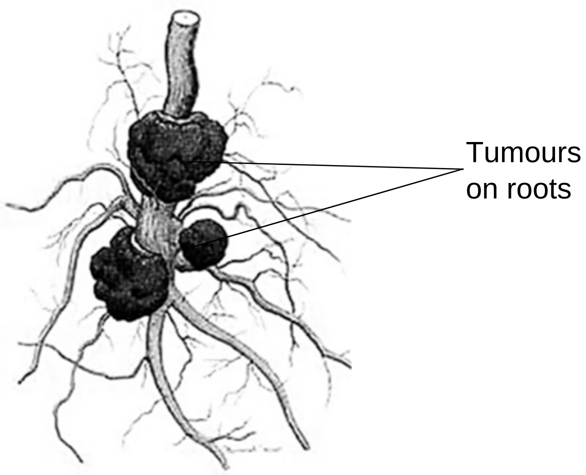

Fig. 2.3

Suggest how the bacteria infection resulted in tumour formation.

1. affects G1 checkpt in interphase

checks for presence of growth factors;

2. auxin regulates gene expression

overpdtn of auxin results in overpdtn of CDKs / cyclins / enz involved in ctrl of cell cycle;

3. auxin activates telomerase

plant cells divide indefinitely;

4. auxin prevents programmed cell death / apoptosis

decreasing activities of prots involved in cell death; [3]

(iii) Suggest why plants with these tumours if left untreated will eventually die.

1. tumours are disorganised tissue with no specialised fns;

2. leading to disruption of normal tpt of essential resources (e.g. sugar / water) for survival; [2]

[Total: 14]

3 In sickle cell anaemia the recessive allele Hb^S replaces the normal allele Hb^A.

- The frequency of Hb^S is much higher in West Africa than in most parts of the world.
- The frequency of Hb^S corresponds with the distribution of malaria.

(a) Explain what is meant by the term *allele*.

1. alternative form of a gene (located at the same gene locus);

2. may be dominant, recessive or codominant; [1]

(b) State whether the likely life expectancy is high or low in West Africa for individuals with the following genotypes. In each case give a reason for your answer.

Hb^AHb^A ***low***

susceptible to / die from, malaria; [1]

Hb^AHb^S ***high***

no / only have traits of sickle cell anaemia, not susceptible to malaria; [1]

Hb^SHb^S ***low***

susceptible to / die from sickle cell anaemia; [1]

(c) Suggest why the frequency of Hb^S allele corresponds to the distribution of malaria.

1. presence of malaria act as selection pressure

selects for indv with favourable phenotype i.e. resistant to malaria;

2. indv with Hb^S allele has (Hb that polymerises thus) shorter RBC lifespan

disrupts life cycle of / kills malaria parasite;

3. Hb^S is a recessive allele

thus heterozygotes do not suffer from effects of SCA & is able to pass down Hb^S allele to offspring (increasing its freq in gene pool); [3]

(d) Describe how the Hb^S allele came about.

1. (spontaneous) gene / point mutation

in β-globin gene;

2. single base sub

of T → A in DNA template strand / GAA → GUA in 6th codon / glu → val; [2]

(e) A man with sickle-cell anaemia marries a woman without sickle-cell anaemia whose mother also has sickle-cell anaemia. Using a genetic diagram, find the probability that the couple would give birth to a son with sickle-cell anaemia.

parent phenotype	***man with sickle cell anaemia***		***woman without sickle cell anaemia***
parent genotype ;	***XY Hb^SHb^S***	***x***	***XX Hb^AHb^S***
gametes ;	***XHb^S YHb^S***		***XHb^A XHb^S***

Punnett Sq ;

genotypic ratio	***$XXHb^AHb^S$:***	***$XXHb^SHb^S$:***	***$XYHb^AHb^S$:***	***$XYHb^SHb^S$***
phenotypic ratio	***1 female normal***	***: 1 female SCA***	***: 1 male normal***	***: 1 male SCA***

Probability = 0.25 ;

[4]

[Total: 13]

4 Reef-building corals are marine invertebrates found in shallow, clear, tropical oceans. The corals secrete an exoskeleton of calcium carbonate that becomes the underlying structure of the coral reef ecosystem.

Zooxanthellae are a group of unicellular algae from the genus *Symbiodinium* that live within the cells of reef-building corals. The relationship has been described as mutualistic since it is beneficial to both coral and zooxanthellae.

(a) Evidence shows that the mutualistic relationship between zooxanthellae and reef-building corals has evolved by free-living algae invading corals that did not contain algae.

(i) Corals that do not need zooxanthellae can live at a greater depth than reef-building corals.

Explain why this is so.

1. corals with no zooxanthellae thus no need light for photosyn

can live at greater depths where less light can penetrate / reach;

2. such corals able to acquire sufficient nutrients thro' own feeding

due to diff feeding method / bcos ocean has more nutrient at greater depths; [2]

(ii) Suggest **two** benefits **to the zooxanthellae** of their association with the corals.

1. corals respire produce CO_2

for zooxanthellae to use in photosynthesis;

2. corals pdc nitrogenous waste / phosphate

as nutrients for zooxanthellae growth / metabolism;

3. corals provide structural support

for zooxanthellae to absorb as much light as possible;

4. provide protection

against predators / harsh environmental conditions; [2]

(b) Under conditions of stress the relationship between the reef-building corals and the zooxanthellae can break down. Loss of zooxanthellae and the subsequent whitening that occurs, shown in Fig. 4.1, is known as coral bleaching. Coral bleaching can lead to death of the coral.

Fig. 4.1

(i) Suggest **two** reasons why permanent loss of zooxanthellae can lead to the death of the coral.

1. decreased source of food / nutrients for corals (pdc by zooxanthellae via photosynthesis)

coral die from starvation / malnutrition;

2. loss of protective algal layer

from harmful effects of direct sunlight; [2]

(ii) Increased sea temperature associated with global climate change is known to be an environmental stress that can cause coral bleaching. The temperature range for healthy survival of reef-building coral is 25°C to 29°C.

Explain why the areas of sea containing coral reefs are susceptible to increased temperature resulting from global climate change.

1. areas of sea where corals are found are usually shallow waters

such areas has larger amts of light & heat penetrating waters → ↑ temp easily (c.f. to deeper waters where less light penetrates);

2. shallow waters also has less water vol

thus absorption of small amts of heat can lead to sig ↑ in temp c.f. to deeper areas; [2]

[Total: 8]

Section B

Answer **one** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a), (b) a**s indicated in the question.

Either

5 **(a)** Stem cells research offers a great number of potential benefits to humans and has helped scientists to understand many cellular processes naturally involved in the growth and development of organisms.

Describe the role of stem cells in humans and discuss the ethical issues that arise from stem cells research. [8]

Role of stem cells

1. ***stem cells have unlimited cell renewal capacity***
   can dy/dx into any specialised cell type;
2. ***zygotic / morula stem cells are totipotent***
   can dy/dx into any cell types including cells of placenta;
3. ***embryonic stem cells (ESCs) are pluripotent***
   can dy/dx into any cell types except cells of placenta;
4. ***ESCs dy/dx into cells of 3 germ layers***
   endoderm, ectoderm, mesoderm;
5. ***adult stem cells are multipotent***
   can dy/dx into specific cell lineages;
6. ***haematopoietic stem cells (HSCs) dy/dx into blood cell components***
   blood cells & platelets;
7. ***HSCs gives rise to white blood cells***
   giving rise to immunity;
8. ***HSCs gives rise to red blood cells***
   tpt gases around the body;

(Max 6m)

Ethical issues

9. ***use of ESCs raises issues of potential loss of a life as embryo destroyed in process & hence denied the chance of developing into a human being;***
10. ***issue of informed consent & relinquished rights by donor over donated tissue e.g. frozen embryos from infertility treatment / gametes;***
11. ***issue of whether tissue was donated based on free choice & implications with paid tissue donation;***
12. ***issue of medical complications in gamete retrieval if donation of gametes (oocytes) occur solely for research purposes;***

13. ***issue of protecting reproductive rights for women involved in infertility treatments, where oocyte of highest quality should be used for reproductive treatment instead of research;***

14. ***confidentiality of donor info could be compromised, resulting in inconveniences / harassment by companies involved in stem cell research;***

(Max 3m)

(b) Outline how genes found on the DNA are expressed to result in the phenotype of an organism. [7]

1. ***genes are nucleotides seq***

   coding for polypeptides / prots / RNA;

2. ***transcription of genes into mRNAs***

   occurs in nucleus;

3. ***RNA pol binds to promoters of specific genes***

   in presence of basal TFs in nucleus;

4. ***(post-transcriptional) modification of mRNA via addition of 5' methylated guanosine cap, poly A tail;***

5. ***(post-transcriptional) modification of mRNA via splicing (by spliceosome) where introns removed & exons joined together into mature mRNA;***

6. ***mRNA translated in cytoplasm***

   results in polypeptide pdtn;

7. ***initiation of translation involves initiator tRNA with met***

   binds to start codon AUG on mRNA;

8. ***ribosome subunits bind to mRNA***

   placing initiator tRNA at P site;

9. ***peptidyl transferase forms peptide bonds***

   b/w amino acids (aas) carried by tRNA on A site & P site;

10. ***1^o seq of aas results in polypeptide folding***

    specific 3D config to perform its specific f^n;

11. ***e.g. of phenotype e.g. sickle cell anaemia;***

[Total: 15]

6 **(a)** Proteins play important roles in the metabolic processes of both plants and animals. In particular, enzymes assist in the catalysis of key activities in the production of organic matter in plants using carbon dioxide as its raw material.

Describe the roles of enzymes in photosynthesis. [9]

(A) light dependent rxn

1. ***<u>Mn-containing enz</u> (H_2O splitting enz) catalyses photolysis of H_2O into H^+, O_2 & e^-;***
2. ***H^+ released into stroma for active tpt across thylakoid memb/ contributes to H^+ grad across thylakoid memb;***
3. ***O_2 evolved as by-pdt;***
4. ***e^- replace e^- deficit in P680 (PSII) to allow non-cyclic photophosphorylation to continue;***
5. ***<u>NADP reductase</u> catalyses reduction of NADP to NADPH (reduced NADP);***
6. ***H^+ from photolysis of H_2O, e^- from ETC;***
7. ***NADPH involved in carbon reduction stage in Calvin cycle;***
8. ***<u>ATP synthase</u> catalyses phosphorylation of ADP with P_i to ATP;***
9. ***with diffusion of H^+ via chemiosmosis;***
10. ***from thylakoid space into stroma;***

(B) Light independent rxn

11. ***<u>Rubisco</u> (RUBP carboxylase-oxygenase) catalyses carbon fixation stage in Calvin cycle;***
12. ***1 mol of CO_2 (1C) is added to 1 mol RUBP (5C);***
13. ***form 6C intermediate then splitted into 2 mols of 3C phosphoglycerate/ glycerate phosphate (PGA);***

Max 3 points per enz. Need to include at least 1 enz from each category

(b) Membranes of organelles in photosynthesis are key in the light dependent stage of photosynthesis. Metabolic poisons are known to disrupt key events in the light dependent stage.

Explain the possible ways metabolic poisons can inhibit light dependent reaction in photosynthesis. [6]

metabolic poisons can block/ prevent activity of prots & enz involved in light dependent rxn

1. ***blocks role of <u>Mn-containing enz</u> (H_2O splitting enz);***
2. ***prevents photolysis of H_2O into H^+, O_2 & e^-;***
3. ***O_2 not released as by-pdt;***
4. ***H^+ not available for reduction of NADP at end of ETC/ no contribution for H^+ grad;***
5. ***e^- not replaced in deficit P680 (PSII);***
6. ***blocks role of <u>pump prot;</u>***
7. ***prevents active tpt of H^+ across thylakoid memb;***

8. ***from stroma into thylakoid space;***
9. ***no generation of H^+ grad for chemiosmosis to take place;***
10. ***blocks role of <u>e^- carriers</u> in ETC;***
11. ***e^- not tpted down energy level;***
12. ***energy not dissipated hence not provided for pumping of H^+ across thylakoid memb;***
13. ***NADP not reduced to NADPH as no e^- supplied by ETC;***
14. ***blocks role of <u>ATP synthase</u>;***
15. ***prevents diffusion of H^+ via chemiosmosis from thylakoid space into stroma;***
16. ***prevents phosphorylation of ADP + P_i into ATP;***

Max 3 points per prot/ enz activity affected. Only need to mention how 2 prot/ enz affected.

[Total: 15]